

An Exact Counterexample to Graffiti 197

AdaIvy project

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Status. This note gives a human-checkable exact counterexample. This result was AI-generated by the AdaIvy project. Its closed arithmetic certificate was kernel-checked in Lean with no axioms; the spectral derivation itself remains human-checkable rather than fully formalized. It creates no AdaIvy epistemic warrant, and novelty, significance, and publication approval remain unassessed.

1 Statement and conventions

The conjecture known as Graffiti 197 states [1, Graffiti 197, PDF p. 71]

$$-\lambda_{\text{second smallest}}(A(G)) \leq \text{range}(\text{spec}(\Gamma(G))),$$

where $A(G)$ is the adjacency matrix and $\Gamma(G)$ is the gravity matrix. This note uses the source-supported convention that the range of a vector is the number of its distinct values.

For a graph with n vertices, degrees $d(u)$, and graph distance $\text{dist}(u, v)$, the gravity matrix is

$$\Gamma_{uv} = \begin{cases} \frac{d(u)d(v)}{(n-1)\text{dist}(u, v)}, & u \neq v, \\ 0, & u = v. \end{cases}$$

Proposition 1. *Graffiti 197 is false under these conventions. The Kneser graph $K(7, 2)$ is a counterexample: its left-hand side is 4, while its gravity matrix has exactly 3 distinct eigenvalues.*

Auditable derivation. The investigation used the Kneser-graph candidate reported by AI Village News [3] as a lead, then independently reconstructed the graph over exact integers and rationals. The argument below counts the strongly regular parameters, derives the adjacency spectrum, transforms it into the gravity spectrum, and compares the two sides. It is a reproducible mathematical account, not a claim to expose private model chain-of-thought.

2 The graph and its adjacency spectrum

The vertices of $K(7, 2)$ are the two-element subsets of a seven-element set; two vertices are adjacent exactly when the corresponding subsets are disjoint. Consequently

$$n = \binom{7}{2} = 21, \quad k = \binom{5}{2} = 10.$$

The graph is strongly regular with parameters

$$(n, k, \lambda, \mu) = (21, 10, 3, 6).$$

Indeed, two adjacent two-subsets use four points and have $\binom{3}{2} = 3$ common neighbors. Two distinct nonadjacent two-subsets share one point, use three points, and have $\binom{4}{2} = 6$ common neighbors. Hence

$$A^2 = (k - \mu)I + (\lambda - \mu)A + \mu J = 4I - 3A + 6J.$$

The all-ones vector has adjacency eigenvalue 10. On its orthogonal complement J vanishes, so every remaining adjacency eigenvalue x obeys

$$x^2 + 3x - 4 = 0, \quad \text{hence} \quad x \in \{1, -4\}.$$

Writing their multiplicities as f and g , dimension and trace give

$$f + g = 20, \quad 10 + f - 4g = 0,$$

and therefore $f = 14$ and $g = 6$. Thus

$$\text{spec}(A) = \{10^1, 1^{14}, (-4)^6\}.$$

Because -4 has multiplicity six, the second-smallest adjacency eigenvalue is -4 . The left-hand side of Graffiti 197 is therefore

$$-\lambda_{\text{second smallest}}(A) = 4.$$

3 The gravity spectrum

The graph has diameter two. Since every degree is 10 and $n - 1 = 20$, distinct adjacent vertices have gravity entry $100/20 = 5$, while distinct nonadjacent vertices have entry $100/40 = 5/2$. Therefore

$$\Gamma = \frac{5}{2}(A + J - I).$$

On the all-ones vector,

$$\Gamma \mathbf{1} = \frac{5}{2}(10 + 21 - 1)\mathbf{1} = 75\mathbf{1}.$$

On the orthogonal complement of $\mathbf{1}$, the matrix J vanishes. The adjacency eigenvalue 1 is sent to

$$\frac{5}{2}(1 - 1) = 0,$$

and the adjacency eigenvalue -4 is sent to

$$\frac{5}{2}(-4 - 1) = -\frac{25}{2}.$$

It follows that

$$\text{spec}(\Gamma) = \left\{ 75^1, 0^{14}, \left(-\frac{25}{2}\right)^6 \right\}.$$

The gravity spectrum has exactly three distinct values. Graffiti 197 would therefore require

$$4 \leq 3,$$

which is false. This completes the counterexample calculation.

4 Verification and scope

An independent Python standard-library checker reconstructed $K(7, 2)$, formed the adjacency and gravity matrices over the rational numbers, computed their characteristic polynomials exactly, and checked the stated factorizations. The checker uses neither floating point nor the downloaded candidate program.

Run disclosure.

AdaIvy’s durable ledger records three completed OpenAI calls to `gpt-5-mini-2025-08-07`. The recorded external model cost is USD 0.017859 against the authorized USD 20 cap. Recorded usage is 7,747 input tokens and 7,960 output tokens, for 15,707 tokens total. This accounting is partial: the surrounding interactive Codex authoring session did not expose billable usage to AdaIvy’s ledger, so its tokens and cost are not included or estimated.

AdaIvy also generated a closed Lean arithmetic certificate covering the four combinatorial counts, the three gravity-eigenvalue transformations, and the final contradiction $\neg(4 \leq 3)$. The sealed Lean 4.32.1 checker accepted it with no axioms. The exact checked source is linked here: `graffiti-197-kneser-counterexample.lean`.

- Exact checker SHA-256:

`2d487616404964fa46306f4ec7a0156e7d267377ee8d0a35280bd0ce4ad39.`

- Exact result content hash:

`sha256:626439796ee4635919cdc3dc8c08f6f7bc13f8928770c7d15b8134905b090a1f.`

- Lean finding: `formal-finding.db32ea12d524eaf9078e94ae`; outcome `kernel_checked`; approved and unapproved axiom lists both empty.

- Lean wrapper SHA-256:

`sha256:24fb637e1b7f06b656392636b85fdc1d5097ee5430c986021721bce0c7fce249.`

- Human-review status: this mathematical result has not yet been approved by a human reviewer in AdaIvy. Automated checks alone do not make it an endorsed result (machine status: `awaiting_human_review`; no warrant created).

The Lean certificate does not prove from first principles that the explicit 21×21 adjacency and gravity matrices have the displayed spectra. It checks the closed exact arithmetic consequences used after that mathematical derivation. A complete matrix-spectrum formalization would require a broader formal representation and approved imports. This result does not settle Graffiti 322. If “range” is instead interpreted as numerical maximum minus minimum, that is a different formulation; Roucairol and Cazenave report a numerical C_{17} counterexample for it [2].

References

- [1] S. Fajtlowicz, *Written on the Wall*, July 2004 compilation, Graffiti 197 on PDF page 71. Source PDF.

- [2] M. Roucairol and T. Cazenave, *Refutation of Spectral Graph Theory Conjectures with Search Algorithms*, arXiv:2409.18626v1 (2024); published in CCIS 2976 (2026). https://doi.org/10.1007/978-3-032-28263-7_2.
- [3] AI Village News, *Opus 5 #124 — WOW 197 gravity range FALSE*, 13 August 2026. Treated as an unreviewed candidate source and independently rechecked here.

*About this project*¹

¹AdaIvy is an open-source AI-assisted mathematical research project that keeps retrieval, computation, formal checking, human review, and publication status as separate recorded boundaries. Source code and reproducibility materials: <https://github.com/jketts/AdaIvy>.